

$i12 \rightarrow$

$$\frac{1}{16} \left(\frac{1}{4} m_1 \tilde{\mu} g_1^4 y_u^{i12} L_{F2,2,0} [m_1, \tilde{\mu}] + \frac{1}{12} m_2 \tilde{\mu} g_2^4 (17 y_u^{i12} - 4 y_u^{p12} \delta_{p11}) L_{F2,2,0} [m_2, \tilde{\mu}] + \right.$$
$$\frac{1}{6} m_2 \tilde{\mu} g_2^4 (-2 y_u^{i12} + y_u^{p12} \delta_{p11}) L_{F3,2,-1} [m_2, \tilde{\mu}] -$$
$$\frac{1}{18} \tilde{\mu} g_1^2 y_d^p \bar{a}_d^p r (3 y_u^{i12} + y_u^{s12} \delta_{s11} - 4 y_u^{i1s} \delta_{s12}) L_{F2,2,0} [m_d^r, m_q^p] +$$
$$\frac{1}{36} \tilde{\mu} g_1^2 y_d^p \bar{a}_d^p r (3 y_u^{i12} + y_u^{s12} \delta_{s11} - 4 y_u^{i1s} \delta_{s12}) L_{F3,2,-1} [m_d^r, m_q^p] -$$
$$\frac{1}{18} \tilde{\mu} g_1^2 y_e^p \bar{a}_e^p r (3 y_u^{i12} + y_u^{s12} \delta_{s11} - 4 y_u^{i1s} \delta_{s12}) L_{F2,2,0} [m_e^r, m_l^p] +$$
$$\frac{1}{36} \tilde{\mu} g_1^2 y_e^p \bar{a}_e^p r (3 y_u^{i12} + y_u^{s12} \delta_{s11} - 4 y_u^{i1s} \delta_{s12}) L_{F3,2,-1} [m_e^r, m_l^p] +$$
$$\frac{1}{18} \tilde{\mu} y_e^p \bar{a}_e^p r (-18 g_2^2 y_u^{i12} + g_1^2 (3 y_u^{i12} + y_u^{s12} \delta_{s11} - 4 y_u^{i1s} \delta_{s12})) L_{F3,1,0} [m_l^p, m_e^r] +$$
$$\frac{1}{18} \tilde{\mu} g_1^2 y_e^p \bar{a}_e^p r (3 y_u^{i12} + y_u^{s12} \delta_{s11} - 4 y_u^{i1s} \delta_{s12}) L_{F3,2,-1} [m_l^p, m_e^r] -$$
$$\frac{1}{6} \tilde{\mu} y_e^p \bar{a}_e^p r (g_2^2 (-19 y_u^{i12} + y_u^{s12} \delta_{s11}) + g_1^2 (3 y_u^{i12} + y_u^{s12} \delta_{s11} - 4 y_u^{i1s} \delta_{s12})) L_{F4,1,-1} [m_l^p, m_e^r] +$$
$$\frac{1}{18} \tilde{\mu} y_e^p \bar{a}_e^p r (3 g_2^2 (-8 y_u^{i12} + y_u^{s12} \delta_{s11}) + g_1^2 (3 y_u^{i12} + y_u^{s12} \delta_{s11} - 4 y_u^{i1s} \delta_{s12}))$$
$$L_{F5,1,-2} [m_l^p, m_e^r] + \frac{1}{18} \tilde{\mu} y_d^p \bar{a}_d^p r (-54 g_2^2 y_u^{i12} + g_1^2 (3 y_u^{i12} + y_u^{s12} \delta_{s11} - 4 y_u^{i1s} \delta_{s12}))$$
$$L_{F3,1,0} [m_d^r, m_q^p] + \frac{1}{18} \tilde{\mu} g_1^2 y_d^p \bar{a}_d^p r (3 y_u^{i12} + y_u^{s12} \delta_{s11} - 4 y_u^{i1s} \delta_{s12}) L_{F3,2,-1} [m_d^r, m_q^p] +$$
$$\frac{1}{24} y_d^p \bar{a}_d^p r (9 g_2^2 (19 y_u^{i12} - y_u^{s12} \delta_{s11}) - 5 g_1^2 (3 y_u^{i12} + y_u^{s12} \delta_{s11} - 4 y_u^{i1s} \delta_{s12}))$$
$$L_{F3,1,-1} [m_q^p, m_d^r] + \frac{1}{6} \tilde{\mu} y_d^p \bar{a}_d^p r$$
$$(3 g_2^2 (-8 y_u^{i12} + y_u^{s12} \delta_{s11}) + g_1^2 (3 y_u^{i12} + y_u^{s12} \delta_{s11} - 4 y_u^{i1s} \delta_{s12})) L_{F5,1,-2} [m_q^p, m_d^r] +$$
$$\frac{1}{36} \tilde{\mu} y_u^p \bar{a}_u^p r (9 g_2^2 (5 y_u^{i12} - y_u^{s12} \delta_{s11}) + g_1^2 (3 y_u^{i12} + y_u^{s12} \delta_{s11} - 4 y_u^{i1s} \delta_{s12}))$$
$$L_{F2,2,0} [m_q^p, m_u^r] + \frac{3}{4} \tilde{\mu} g_2^2 y_u^p r y_u^{i12} \bar{a}_u^p r L_{F3,1,0} [m_q^p, m_u^r] -$$
$$\frac{1}{72} \tilde{\mu} y_u^p \bar{a}_u^p r (9 g_2^2 (5 y_u^{i12} - y_u^{s12} \delta_{s11}) + g_1^2 (3 y_u^{i12} + y_u^{s12} \delta_{s11} - 4 y_u^{i1s} \delta_{s12}))$$
$$L_{F3,2,-1} [m_q^p, m_u^r] - \frac{3}{4} \tilde{\mu} g_2^2 y_u^p r y_u^{i12} \bar{a}_u^p r L_{F4,1,-1} [m_q^p, m_u^r] - \frac{1}{36} \tilde{\mu} y_u^p \bar{a}_u^p r$$
$$(9 g_2^2 (8 y_u^{i12} - y_u^{s12} \delta_{s11}) + g_1^2 (3 y_u^{i12} + y_u^{s12} \delta_{s11} - 4 y_u^{i1s} \delta_{s12})) L_{F3,1,0} [m_u^r, m_q^p] -$$
$$\frac{1}{36} \tilde{\mu} y_u^p \bar{a}_u^p r (9 g_2^2 (5 y_u^{i12} - y_u^{s12} \delta_{s11}) + g_1^2 (3 y_u^{i12} + y_u^{s12} \delta_{s11} - 4 y_u^{i1s} \delta_{s12}))$$
$$L_{F3,2,-1} [m_u^r, m_q^p] - \frac{1}{12} \tilde{\mu} y_u^p \bar{a}_u^p r$$
$$(9 g_2^2 (-8 y_u^{i12} + y_u^{s12} \delta_{s11}) + g_1^2 (3 y_u^{i12} + y_u^{s12} \delta_{s11} - 4 y_u^{i1s} \delta_{s12})) L_{F4,1,-1} [m_u^r, m_q^p] +$$
$$\frac{1}{6} \tilde{\mu} y_u^p \bar{a}_u^p r (3 g_2^2 (-8 y_u^{i12} + y_u^{s12} \delta_{s11}) + g_1^2 (3 y_u^{i12} + y_u^{s12} \delta_{s11} - 4 y_u^{i1s} \delta_{s12}))$$
$$L_{F5,1,-2} [m_u^r, m_q^p] - \frac{1}{2} m_1 \tilde{\mu} g_1^2 g_2^2 y_u^{i12} L_{F3,1,0} [\tilde{\mu}, m_1] - \frac{7}{8} m_1 \tilde{\mu} g_1^4 y_u^{i12} L_{F3,2,-1} [\tilde{\mu}, m_1] -$$
$$\frac{1}{24} m_1 \tilde{\mu} g_1^2 (3 g_2^2 (-14 y_u^{i12} + y_u^{p12} \delta_{p11}) + g_1^2 (3 y_u^{i12} + y_u^{p12} \delta_{p11} - 4 y_u^{i1p} \delta_{p12}))$$
$$L_{F4,1,-1} [\tilde{\mu}, m_1] + \frac{1}{2} m_1 \tilde{\mu} g_1^4 y_u^{i12} L_{F4,2,-2} [\tilde{\mu}, m_1] +$$
$$\frac{1}{18} m_1 \tilde{\mu} g_1^2 (3 g_2^2 (-8 y_u^{i12} + y_u^{p12} \delta_{p11}) + g_1^2 (3 y_u^{i12} + y_u^{p12} \delta_{p11} - 4 y_u^{i1p} \delta_{p12}))$$
$$L_{F5,1,-2} [\tilde{\mu}, m_1] + \frac{1}{6} m_2 \tilde{\mu} g_2^4 (-13 y_u^{i12} + 2 y_u^{p12} \delta_{p11}) L_{F3,1,0} [\tilde{\mu}, m_2] +$$
$$\frac{1}{24} m_2 \tilde{\mu} g_2^4 (-73 y_u^{i12} + 8 y_u^{p12} \delta_{p11}) L_{F3,2,-1} [\tilde{\mu}, m_2] +$$
$$\frac{1}{6} m_2 \tilde{\mu} g_2^2 (g_2^2 (50 y_u^{i12} - 7 y_u^{p12} \delta_{p11}) - g_1^2 (3 y_u^{i12} + y_u^{p12} \delta_{p11} - 4 y_u^{i1p} \delta_{p12}))$$
$$L_{F4,1,-1} [\tilde{\mu}, m_2] + 2 m_2 \tilde{\mu} g_2^4 y_u^{i12} L_{F4,2,-2} [\tilde{\mu}, m_2] +$$
$$\frac{1}{2} m_2 \tilde{\mu} g_2^2 (3 g_2^2 (-8 y_u^{i12} + y_u^{p12} \delta_{p11}) + g_1^2 (3 y_u^{i12} + y_u^{p12} \delta_{p11} - 4 y_u^{i1p} \delta_{p12}))$$
$$L_{F5,1,-2} [\tilde{\mu}, m_2] - \frac{1}{36} m_1 \tilde{\mu} g_1^2 \bar{y}_d^p r y_d^{i1r} y_u^{p12} L_{F2,1,1,0} [m_1, m_d^r, m_q^{i1}] -$$
$$\frac{1}{72} m_1 \tilde{\mu} g_1^2 \bar{y}_d^p r y_d^{i1r} y_u^{p12} L_{F2,2,1,-1} [m_1, m_d^r, m_q^{i1}] +$$
$$\frac{1}{36} m_1 \tilde{\mu} g_1^2 \bar{y}_d^p r y_d^{i1r} y_u^{p12} L_{F3,1,1,-1} [m_1, m_d^r, m_q^{i1}] - \frac{1}{6} m_1 \tilde{\mu} g_1^2 \bar{y}_d^p r y_d^{i1r}$$
$$y_u^{p12} L_{F2,1,1,0} [m_1, m_d^r, \tilde{\mu}] + \frac{1}{6} m_1 \tilde{\mu} g_1^2 \bar{y}_d^p r y_d^{i1r} y_u^{p12} L_{F3,1,1,-1} [m_1, m_d^r, \tilde{\mu}] +$$
$$\frac{1}{18} m_1 \tilde{\mu} g_1^2 \bar{y}_d^p r y_u^{p12} y_u^{i1r} L_{F2,1,1,0} [m_1, m_q^p, m_u^{i2}] + \frac{1}{36} m_1 \tilde{\mu} g_1^2 \bar{y}_d^p r y_u^{p12} y_u^{i1r}$$
$$L_{F2,2,1,-1} [m_1, m_q^p, m_u^{i2}] - \frac{1}{18} m_1 \tilde{\mu} g_1^2 \bar{y}_d^p r y_u^{p12} y_u^{i1r} L_{F3,1,1,-1} [m_1, m_q^p, m_u^{i2}] +$$
$$\frac{1}{12} m_1 \tilde{\mu} g_1^2 \bar{y}_d^p r y_u^{p12} y_u^{i1r}$$